



Kishoreganj University
3rd Year 2nd Semester B.Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3201 : Computer Networking (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.
Answer 05 out of 07 questions.

- ✓ 1. (a) Describe the layers of the Internet protocol stack and mention the function of each layer. And compare it to the OSI reference model. 5
(b) A router needs to transmit a packet of length $L = 7.5$ Mbits over a link with a transmission rate of $R = 1.5$ Mbps. The physical link length is $d = 1,000$ km, and the propagation speed is $s = 2 \times 10^8$ m/sec. 5
 - i) Calculate the transmission delay for this packet.
 - ii) Calculate the propagation delay.
 - iii) Explain the difference between these two types of delay based on your results.
- (c) Differentiate between packet switching and circuit switching. Discuss their respective advantages, particularly focusing on how packet switching handles bursty data traffic and resource sharing. 4
- ✓ 2. (a) Describe the working steps of sending an email from sender to receiver. 5
(b) Explain the working of DNS resolution using iterative query. Also discuss the role of caching. 5
(c) Compare FDM and TDM in terms of efficiency and usage in modern networks. 4
3. (a) Give an example scenario to send a message using SMTP protocol. 5
(b) Explain the steps involved in "A day in the life of a web request" from typing a URL to receiving the web page. 5
(c) A company decides to replace Telnet with SSH and implement DHCP for all employees. What are the possible challenges? 4
4. (a) What is the importance of subnetting in large organizations? 3
(b) Given the network address 132.21.0.0, find the class, the block, and the range of the addresses. 3
(c) An organization is granted the block 130.34.12.64/26. The organization needs to have four subnets. 6
 - i) What is the subnet prefix length?
 - ii) What are the subnet addresses and the range of addresses for each subnet?
 - iii) Design the subnets.
- (d) You are assigned an IP address of 172.30.0.0 and you need 1000 hosts on your network, what is your subnet mask? 2
5. (a) What is convergence? Why is slow convergence a problem in DVR? Explain in detail with a figure. 3
(b) Consider a network with nodes A, B, C, and D. The link costs are: A-B = 2, A-C = 5, 5

B-C = 2, B-D = 1, C-D = 3.

Use Dijkstra's algorithm to find the shortest path from node A to all other nodes. Show the steps (iterations) of the algorithm.

- (c) Compare and contrast the Distance Vector (DV) and Link-State (LS) routing algorithms based on three criteria: message complexity, speed of convergence, and robustness. 4
6. (a) If the data link layer is reliable and has flow and error control, why do we need these functions at the transport layer as well? 2
- (b) Consider the following figure, Show the forwarding process if a packet arrives at R1 in Figure-1 with the destination address 201.4.22.35. 4

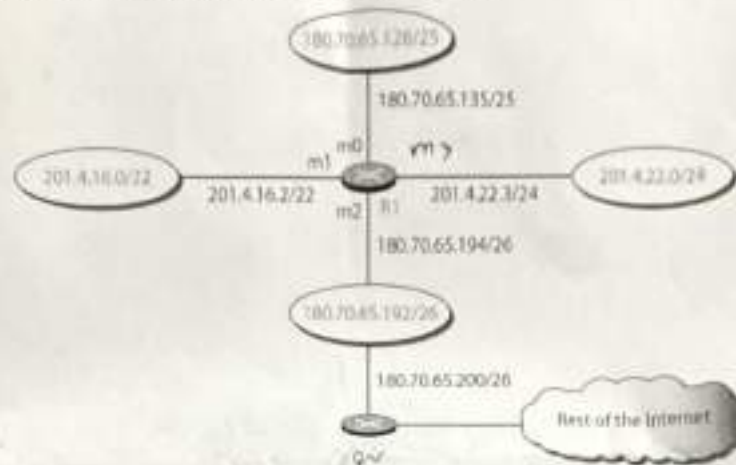


Figure-1

- (c) What is Network Address Translation (NAT)? Discuss its primary motivation and explain how it uses a translation table to manage private and public IP addresses. 5
- (d) Define wireless communication and mobility. 3
7. (a) What is CSMA/CA? Explain how CDMA allows multiple users to transmit simultaneously. 5
- (b) Consider a line graph showing a TCP sender's Congestion Window Size (cwnd) on the Y-axis versus Transmission Rounds on the X-axis. The graph starts at 1, rises exponentially to 16, then rises linearly to 24. At round 12, it drops sharply to 12 and rises linearly again to 20. At round 18, it drops all the way to 1 and starts rising exponentially again. Based on the TCP Congestion Window graph provided: 5
- Identify the transmission rounds where TCP is operating in the Slow Start phase.
 - Identify the transmission rounds where TCP is operating in the Congestion Avoidance phase.
 - At round 12, a segment loss was detected. Was it detected via a Timeout or Three Duplicate ACKs? Justify your answer based on how the graph behaves.
 - What is the value of the slow-start threshold (ssthresh) at the 15th transmission round?
 - At round 18, a segment loss is detected again. Based on the graph, what event caused this loss, and what is the new ssthresh value immediately after?
- (c) Describe error control mechanism of TCP in case of corrupted segment. 4